

IIT JAM Mathematics

Differential Equations Cheat Sheet

1. First-Order ODEs

Method of Separation of Variables

Form: $\frac{dy}{dx} = f(x)g(y)$

Solution: Separate variables and integrate:

$$\int \frac{dy}{g(y)} = \int f(x)dx + C$$

Homogeneous Differential Equations

Form: $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ or $M(x, y)dx + N(x, y)dy = 0$ where M, N are homogeneous of the same degree.

Method: Substitute $y = vx$. Then $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

Equation reduces to variables separable:

$$\frac{dv}{f(v)-v} = \frac{dx}{x}$$

Exact Differential Equations

Form: $M(x, y)dx + N(x, y)dy = 0$

Condition for Exactness: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution:

$$\int_{y \text{ const}} M dx + \int (\text{terms of } N \text{ free from } x) dy = C$$

Integrating Factors (I.F.)

If $Mdx + Ndy = 0$ is not exact, multiply by an I.F. to make it exact.

1. **Function of x only:** If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$,

then I.F. $= e^{\int f(x)dx}$.

2. **Function of y only:** If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = g(y)$,

then I.F. $= e^{\int g(y)dy}$.

3. **Homogeneous:** If eq is homogeneous and $Mx + Ny \neq 0$, I.F. $= \frac{1}{Mx+Ny}$.

4. **Specific Form:** If $yf_1(xy)dx + xf_2(xy)dy = 0$ and $Mx - Ny \neq 0$, I.F. $= \frac{1}{Mx-Ny}$.

Linear First-Order Equations

Form: $\frac{dy}{dx} + P(x)y = Q(x)$

I.F. $= e^{\int P(x)dx}$

Solution: $y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})dx + C$

Bernoulli's Equation

Form: $\frac{dy}{dx} + P(x)y = Q(x)y^n$ ($n \neq 0, 1$)

Method: Divide by y^n : $y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x)$.

Substitute $v = y^{1-n} \implies \frac{dv}{dx} = (1-n)y^{-n}\frac{dy}{dx}$.

Equation reduces to a standard linear form:

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

2. Applications of First-Order

Orthogonal Trajectories

A family of curves that intersects every curve of a given family at right angles.

Cartesian Coordinates: $f(x, y, c) = 0$

1. Differentiate to find ODE: $F(x, y, \frac{dy}{dx}) = 0$.

2. Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$.

3. Solve the new ODE for orthogonal trajectories.

Polar Coordinates: $f(r, \theta, c) = 0$

1. Differentiate to find ODE in $r, \theta, \frac{dr}{d\theta}$.

2. Replace $\frac{dr}{d\theta}$ with $-r^2 \frac{d\theta}{dr}$.

3. Solve the new ODE.

3. Second-Order Linear ODEs

General Form: $y'' + P(x)y' + Q(x)y = R(x)$

Linear ODEs with Constant Coefficients

Form: $y'' + ay' + by = R(x)$ (where $a, b \in \mathbb{R}$)

Complementary Function (C.F.): y_c :

Solve Auxiliary Equation (A.E.): $m^2 + am + b = 0$.

- **Real & Distinct** (m_1, m_2): $y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

- **Real & Equal** ($m_1 = m_2 = m$): $y_c = (c_1 + c_2 x) e^{mx}$

- **Complex** ($\alpha \pm i\beta$): $y_c = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Particular Integral (P.I.): y_p : $y_p = \frac{1}{f(D)} R(x)$ where $D = \frac{d}{dx}$.

- $R(x) = e^{kx}$: $P.I. = \frac{1}{f(k)} e^{kx}$ (if $f(k) \neq 0$).

If $f(k) = 0$, $P.I. = x \frac{1}{f'(k)} e^{kx}$.

- $R(x) = \sin(kx)$ or $\cos(kx)$: Replace D^2 with $-k^2$ (if $f(-k^2) \neq 0$).

- $R(x) = x^k$: Expand $[f(D)]^{-1}$ using binomial theorem.

- $R(x) = e^{kx} V(x)$: $P.I. = e^{kx} \frac{1}{f(D+k)} V(x)$.

- $R(x) = x V(x)$: $P.I. = \left[x - \frac{f'(D)}{f(D)} \right] \frac{1}{f(D)} V(x)$.

General Solution: $y = y_c + y_p$

Method of Variation of Parameters

Used to find P.I. when C.F. is known. For $y'' + P(x)y' + Q(x)y = R(x)$:

Let C.F. be $y_c = c_1 u_1(x) + c_2 u_2(x)$.

Compute Wronskian $W = u_1 u_2' - u_2 u_1'$. ($W \neq 0$)

The Particular Integral is $y_p = A(x)u_1(x) + B(x)u_2(x)$ where:

$$A(x) = - \int \frac{u_2 R(x)}{W} dx, \quad B(x) = \int \frac{u_1 R(x)}{W} dx$$

Cauchy-Euler Equation

Form: $x^2 y'' + pxy' + qy = R(x)$ (constants p, q)

Method: Substitute $x = e^z \implies z = \ln x$ (for $x > 0$).

Let $D \equiv \frac{d}{dz}$. Then:

- $x \frac{dy}{dx} = Dy$

- $x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

Substitute to get a linear ODE with constant coefficients:

$$[D(D-1) + pD + q]y = R(e^z)$$

Solve for $y(z)$, find C.F. and P.I. in terms of z , then substitute $z = \ln x$ back for final solution.